

## ***DIY Activity: Customer Feedback and Service Efficiency Analysis***

**Guidelines for Students:** This is a self-paced Do-It-Yourself (DIY) activity and **submissions are not required**. The primary objective is to enhance your skills through practical, hands-on learning. Engage in discussions with peers to deepen your understanding and collaboratively work through the activity. If you have any questions or need clarification, **please raise them during the upcoming Live Sessions** for a more comprehensive explanation.

**Overview:** Customer satisfaction is being impacted by product quality issues and inefficient support resolution times, leading to negative feedback. Understanding the relationship between feedback scores, resolution times, and product categories is crucial for improving service efficiency, product reliability, and overall customer experience.

### **Dataset:**

#### **[customer\\_feedback\\_dataset](#)**

- **Customer ID** - Unique identifier for each customer.
- **Product Category** - The category of the purchased product.
- **Purchase Date** - The date when the product was purchased.
- **Feedback Score** - Customer's rating of the product (1-10).
- **Comments** - Text feedback provided by the customer.
- **Resolution Time** - Time taken (minutes) to resolve the customer's issue.
- **Support Agent ID** - Unique identifier for the support agent handling the case.

### **Tasks to be performed:**

- Analyze the distribution of 'feedback score' across all product categories. Identify which product category has the highest percentage of negative feedback (scores 1-3) and the lowest percentage of positive feedback (scores 8-10).
- Using a pivot table, calculate the average resolution time for each product category. Identify which category has the longest average resolution time and suggest reasons based on patterns in the dataset. From the same or a second pivot table, also identify which category has the shortest average resolution time (best-performing category).

Separately, using another pivot table, identify which product category has the highest volume of feedback scores rated 9 or 10.

*(Hint: Use PIVOT TABLES — one grouping Product Category by Average Resolution Time, and a second grouping Product Category by count of feedback scores 9–10.)*

- *Analyze the dataset using a pivot table to determine whether there is a seasonal trend in feedback scores by adding rows for each month and calculating the average feedback scores for those months. Specifically, identify which months have the highest and lowest average feedback scores to uncover any patterns or trends throughout the year.*
- *For feedback scores rated 1 or 2, calculate the average resolution time and compare it to the overall average. Does the resolution time improve for extremely dissatisfied customers, or is it consistent across feedback scores?*  
*(Hint: The use of the AVERAGEIF function could be beneficial.)*
- *Determine if ticket volume is evenly distributed across product categories. Calculate the number of tickets per category, identify any significant disparities, and suggest reasons for the imbalance.*  
*(Hint: Use COUNTIF or a PIVOT TABLE to count tickets per Product Category.)*
- *Calculate the percentage of feedback entries that include comments. Identify whether certain product categories are more likely to receive comments in customer feedback.*  
*(Hint:*
  - ➔ *Use the ISBLANK function to check whether comments are present in each row.*
  - ➔ *Then, create a pivot table to visualize the comment rate in relation to product category.)*
- *Investigate whether resolution times tend to decrease with higher feedback scores. Create a trend analysis to determine if faster resolution times correlate with higher customer satisfaction.*  
*(Hint: There is a usage of PIVOT TABLE to visualize the trend by calculating the Average Resolution Time for Each Feedback Score.)*